Dong Wang

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Education

- Ph.D., Electrical and Computer Engineering — 2003
  Carnegie Mellon University, Pittsburgh, PA
- M.S., Electrical and Computer Engineering — 1999
  Carnegie Mellon University, Pittsburgh, PA
- B.S., Computer Science and Engineering — 1995
  Beijing Institute of Technology, Beijing, China

Work Experiences

Senior Staff, AI Research Engineer, **Meta**, Menlo Park, CA — 11/2018-present

FAIR Researcher - Tech lead on Horizontal Data Team (2024-present)

- WebDev RL for Avocado post-training: Work with TBD Lab to drive WebDev skills for Meta personal super-agent, integrating WebDev environments into Avocado eval and RL pipeline, improving build success and pairwise grader rewards through RL.
- Lead Project MATRIX for synthetic data generation: Design and develop a decentralized, peer-to-peer multi-agent framework for synthetic data generation, achieving tens of thousands of concurrent workflows with modular architecture to support diverse use cases. MATRIX is open-sourced in the FAIR 2025Q2 bundle and published at MLSys 2026 GitHub.
- Drive Cross-Functional AI Research Collaborations on post-training: Enable advanced reasoning capacities through the Coral collaborative reasoner, NaturalReasoning data curation and NaturalThought reasoning data selection. Explore multi-turn multi-agent RL for mental world modeling GitHub.
- Vision foundation model data and pre-training: Drive the data curation to deliver a 2.3B image/alt-text dataset for Meta CLIP 2 achieving SOTA multilingual benchmarks. Dataset is adopted by Llama vision foundation models. Develop media compliance mitigation pipeline for many FAIR research teams. Work on Masked Autoencoder for video SSL: 3D ROPE embeddings and Muon optimizer.

ML Engineer - Tech lead on Ranking ML Infra (before 2024)

- Lead Facebook Feature Framework (F3) unification for Messenger ML Infra: Migrate streaming feature pipelines to F3, optimize C++ serving platform to enable Koski-based feature serving in production, achieving CPRA compliance using privacy aware infra (PAI).
- Co-lead the on-device ML platform for Facebook Apps: Design and scale the Mantle on-device platform, lead collaborations for adoption across Meta's Family of Apps (Messenger, FB Stories, FB Dating).

Chief Architect, Deep Learning for Computer Vision, **Matroid Inc.**, Palo Alto, CA — 06/2016-11/2018

- Facial characteristic modeling: Train age gender Keras CNN classification model with 1.4 million parameters. Clean and create balanced dataset with 33k faces from age 1 to 70. Gender accuracy 97.5%, age MAE 4.9 years. Model runs under 250ms on a mobile camera and is also integrated in an Android App using Tensorflow Mobile.
- Tensorflow model serving: Implement a C++ GRPC server using Tensorflow serving to dynamically load and serve models. Integrate MTCNN face detection and alignment. Debug production issues, eg GPU multiple session bug GitHub.
- Realtime task scheduler: Implement a fault tolerant distributed real time scheduler in Python to support model training, serving and video detection tasks. Match task resource requirement with worker resources. Support task dependencies as a DAG. Dynamically scale up and down workers through Kubernetes autoscaler on AWS.

Research Scientist, Machine Learning for e-commerce, **Houzz Inc.**, Palo Alto, CA — 12/2015-05/2016

- Integrate Spark with Luigi job scheduler in YARN.

Software Engineer, Cloud infrastructure, **Databricks Inc.**, San Francisco, CA 06/2014-12/2015

- Spark as a Service: Work on Spark cluster manager on AWS implementing resource allocation, health monitoring, auto scaling, security and 3rd party integration.
- OSS contribution: Implement async execution and cancellation for Spark SQL JDBC thrift server. GitHub.

Senior Software Engineer, Search and Relevance, **Twitter Inc.**, San Francisco, CA 11/2009-06/2014

- User recommendation: Implement cosine similarity computation for >100M Twitter users on Hadoop using weighted sampling and two-hop random walk. Create ML models for user similarity prediction.
- Scaling production services: Design and implement services for who to follow, related tweets, search top tweets.

Senior Software Engineer, Display Advertising, **Yahoo! Inc.**, Santa Clara, CA 06/2008-10/2009

- Rightmedia exchange Ad serving and a real-time Ads bidding engine.

Staff Engineer, Advanced Technology Group, **Synopsys Inc.**, Mountain View, CA 09/2004-05/2008

- Formal verification based on symbolic simulation.

Search Technologist, Model Checking, **0-In Design Automation**, San Jose, CA 05/2003-08/2004

- Build Model checker based on Boolean satisfiability to find logic bugs in ASIC designs.

SELECTED PUBLICATIONS

- L. Yang, S. W. Li, Y. Li, X. Lei, **D. Wang**, A. Mohamed, H. Zhao and H. Xu, "In Pursuit of Pixel Supervision for Visual Pre-training", arXiv:2512.15715, CVPR, 2026.
- **D. Wang**, Y. Li, A. Ni, C. F. Yeh, Y. Emad, X. Lei, L. Robbins, K. Padthe, H. Xu, X. Li, A. Celikyilmaz, R. Raghavendra, L. Huang, C. Wu and S. Li, "Matrix: Peer-to-Peer Multi-Agent Synthetic Data Generation Framework", arXiv:2511.21686, MLSys, 2026.
- Y. S. Chuang, Y. Li, **D. Wang**, C. F. Yeh, K. Lyu, R. Raghavendra, J. Glass, L. Huang, J. Weston, L. Zettlemoyer, X. Chen, Z. Liu, S. Xie, W. Yih, S. Li and H. Xu, "Meta CLIP 2: A Worldwide Scaling Recipe", arXiv:2507.22062, NeurIPS, 2025.
- Y. Li, Y. Emad, K. Padthe, J. Lanchantin, W. Yuan, T. Nguyen, J. Weston, S. Li, **D. Wang**, I. Kulikov and X. Li, "NaturalThoughts: Selecting and Distilling Reasoning Traces for General Reasoning Tasks", arXiv:2507.01921, 2025.
- W. Yuan, J. Yu, S. Jiang, K. Padthe, Y. Li, I. Kulikov, K. Cho, **D. Wang**, Y. Tian, J. E. Weston and X. Li, "NaturalReasoning: Reasoning in the Wild with 2.8M Challenging Questions", arXiv:2502.13124, NeurIPS, 2025.
- A. Ni, R. Desai, Y. Li, X. Lei, **D. Wang**, J. Zhang, J. Yu, R. Raghavendra, G. Ghosh, S. Li and A. Celikyilmaz, "Collaborative Reasoner: Self-Improving Social Agents with Synthetic Conversations", NeurIPS, 2025.
- K. Kamath, A. Sharma, **D. Wang** and Z. Yin, "RealGraph: User Interaction Prediction at Twitter", 2nd workshop on User Engagement Optimization, KDD 2014.
- A. Goel, A. Sharma, **D. Wang** and Z. Yin, "Discovering Similar Users on Twitter", 11th workshop on Mining and Learning with Graphs, KDD 2013.
- P. Gupta, A. Goel, J. Lin, A. Sharma, **D. Wang**, R. Zadeh, "WTF: The Who To Follow Service at Twitter", 22nd International World Wide Web Conference, WWW 2013.
- **D. Wang** and J. Levitt, "Automatic Assume Guarantee Analysis for Assertion-Based Formal Verification", Proceedings of the 2005 Asia and South Pacific Design Automation Conference (ASP-DAC), 2005.
- E. M. Clarke, O. Grumberg, M. Talupur, **D. Wang**, "Making Predicate Abstraction Efficient: How to Eliminate Redundant Predicates", 15th International conference on Computer Aided Verification (CAV), 2003.
- P. Chauhan, E. Clarke, J. Kukula, S. Sapra, H. Veith and **D. Wang**, "Automated Abstraction Refinement for Model Checking Large State Spaces Using SAT Based Conflict Analysis", International Conference on Formal Methods in Computer-Aided Design (FMCAD), 2002.
- **D. Wang**, P. H. Ho, J. Long, J. Kukula, Y. Zhu, T. Ma and R. Damiano, "Formal Property Verification by Abstraction Refinement with Formal, Simulation and Hybrid Engines", Proceedings of the 38th Annual Design Automation Conference (DAC), 2001.